

Mini Project 2

Project Description

You are developing a program that converts English sentences into predicate logic expressions and vice versa. The program will support quantifiers and allow users to input statements in both English and predicate logic notation.

Tasks:

1- English Sentences to Predicate Conversion:

Implement a function that converts English sentences into predicate logic expressions.

Support quantifiers like "all," "some," "no," etc., and map them to $\forall x$, $\exists x$, $\neg\forall x$, $\neg\exists x$, etc., respectively.

2- Predicate to English Sentences Conversion:

Implement a function that converts predicate logic expressions into English sentences.

Support quantifiers like $\forall x$, $\exists x$, $\neg\forall x$, $\neg\exists x$, etc., and map them to "all," "some," "no," etc., respectively.

Example Conversion Tasks:

1- English to Predicate Conversion:

Input: "All dinosaurs are extinct."

Output: " $\forall x (is_dinosaur(x) \rightarrow is_extinct(x))$ "

2- Predicate to English Conversion:

Input: " $\exists x (is_animal(x) \wedge is_endangered(x))$ "

Output: "Some animals are endangered."

Hint:

- 1- Use these words to express quantifiers (**All, every, each, no, none, some, a, an.**)
- 2- Use only these predicates (**are: $\rightarrow$, is: $\rightarrow$, are not: $\rightarrow$, is not: $\rightarrow$**)
- 3- You can use logical connectors like and, OR

Note: Show at least one example of testing the validity of both functions.